For these exercises, you should read the notes on Recurrent Neural Networks.

1) Representing sequences

We are interested in making a neural network that can predict the next element in a sequence. For each of the sequences below, indicate *all* of the following models that could make correct predictions if its weights were set correctly (don't worry about learning). The possible models are:

- A **bigram** model, which is a feedforward network that takes sequence element y_{t-1} as input and generates element y_t as output.
- A **trigram** model, which is a a feedforward network that takes sequence elements y_{t-2} and y_{t-1} as input and generates element y_t as output.
- A simple **recurrent** model, with a one-dimensional state, governed by the equations: $s_t = \tanh(W^{s,s}s_{t-1} + W^{s,x}y_{t-1})$ and $P(y_t = k \mid y_0, \dots y_{t-1}) = \mathrm{softmax}(W^O s_t)$.

1A)

$1, 1, -1, 1, 1, -1, 1, 1, -1, \dots$

- [ ] bigram
- [x] trigram
- [x] recurrent

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1B)

$1, -1, 1, -1, 1, -1, \dots$

- [x] bigram
- [x] trigram
- [x] recurrent

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1C)

$1, 1, 1, -1, 1, 1, 1, -1, \ldots$

- [ ] bigram
- [ ] trigram
- [x] recurrent

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2) RNN Learning

Consider a very simple RNN, defined by the following equations:

$$s_t = ws_{t-1} + x_t$$
$$p_t = s_t$$

There is a single scalar parameter w. Assume that $s_0 = 0$. Note that p_t is the actual output of the RNN at time t, and we want this to equal the corresponding true label y_t.

2A) For an RNN that, given $x = [1, 1, 1]$ produces $p = [1, 2, 3]$, what is the value of w?

Enter a number: 1

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2B) Assume that the loss at a single time step is $\text{Loss}(p_t, y_t) = \frac{1}{2}(p_t - y_t)^2$. Recall p_t is the actual output and y_t is the target output. What is $\frac{d\text{Loss}}{dw}$ where $\text{Loss} = \sum_{t=1}^{3} \text{Loss}(p_t, y_t)$? Assume $s_0 = 0$.

Enter an expression in terms of (only) `w`, `x_1`, `x_2`, `x_3`, `y_1`, `y_2`, `y_3`. Do not use any of the p_t.

dLoss/dw = (x_1*w + x_2 - y_2)*x_1 + (x_1*w**2 + x_2*w + x_3 - y_3)*(2*x_1*w

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Solution: `(x_1*w + x_2 - y_2)*x_1 + (x_1*w**2 + x_2*w + x_3 - y_3)*(2*x_1*w + x_2)`

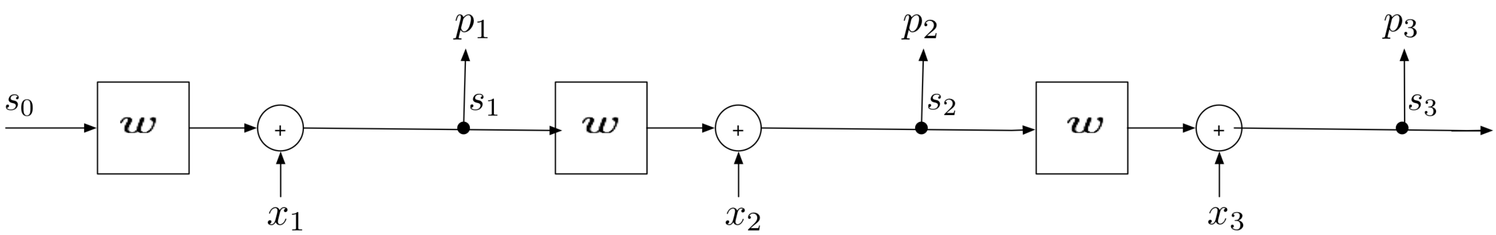
$$(x1 \times w + x2 - y2) \times x1 + \left(x1 \times w^2 + x2 \times w + x3 - y3\right) \times (2 \times x1 \times w + x2)$$

Explanation:

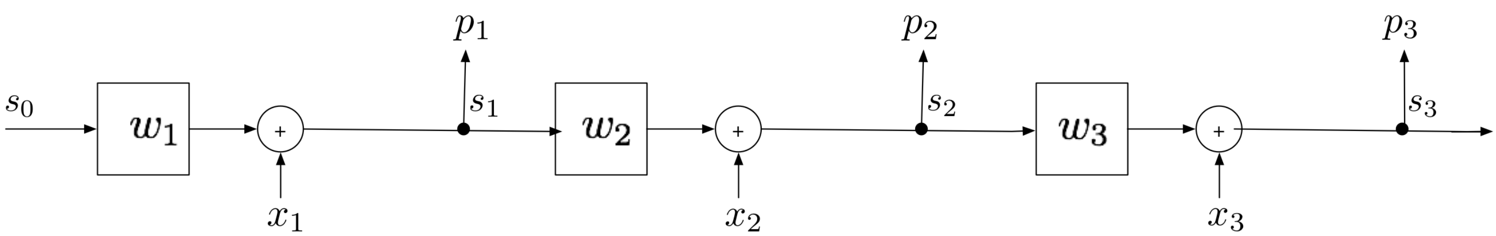
We can unroll the RNN equations to obtain $p1 = x1$, $p2 = wx1 + x2$, and $p3 = w^2 x1 + wx2 + x3$. Thus $\frac{\partial p1}{\partial w} = 0$, $\frac{\partial p2}{\partial w} = x1$, and $\frac{\partial p3}{\partial w} = 2wx1 + x2$. Finally, $\frac{d\text{Loss}}{dw} = \sum_{t=1}^{3}(pt - yt)\frac{\partial pt}{\partial w}$, and we just need to plug in the expressions.

3) Weights

Here is the unrolled RNN from the previous problem. Let's call this the *single-w model*



Consider in addition a related model where instead of a single weight w, at each time step we can use a different weight w_t; call this the *multi-w model*.



Assume $s_0 = 0$ throughout.

3A) Can the single-w model exactly represent a machine that takes $x = [1, 1, 1]$ and produces $y = [1, 2, 4]$? If so, what is the value of w? If not, enter `'None'`.

Enter a number or `'None'`:

'None'

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3B) Can the multi-w model exactly represent a machine that takes $x = [1, 1, 1]$ and produces $y = [1, 2, 4]$? If so, what are the values of $[w1, w2, w3]$ as a Python list? If not, enter `'None'`.

Enter a list of 3 numbers or `'None'`:

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3C) Which of the following are reasons to prefer the single-w model (over the multi-w model)? Note when we say ``memorize a training sequence $[x1, x2, \ldots, xn]$" we mean that the model can potentially produce any output sequence $[y1, y2, \ldots, yn] \in \mathbb{R}^n$ given this input.

Choose all that are true:

- [ ] It can represent a bigger class of sequences.
- [ ] It will generalize better with a small amount of data.
- [ ] It will capture the data's structure as a finite state machine.
- [ ] It can memorize any single training sequence

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3D) Which of the following are reasons to prefer the multi-w model (over the single-w model)?

Choose all that are true:

- [ ] It can represent a bigger class of sequences.
- [ ] It will generalize better with a small amount of data.
- [ ] It will capture the data's structure as a finite state machine.
- [ ] It can memorize any single training sequence

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